

MINERALS AND ENERGY RESOURCES

- A naturally occurring substance that has a definite chemical composition and physical properties is called mineral.
- Minerals are formed in different types of geological environments, under varying conditions. They are created by natural processes without any human interference.

Minerals	Countries in Decreasing Order of Production
Iron ore	China, Australia, Brazil, India, Russia
Manganese	China, South Africa, Australia, Brazil, Gabon
Bauxite	Australia, China, Brazil
Aluminium	China, Russia, Canada, UAE, India
Copper	Chile, China, Peru, USA, Congo
Silver	Mexico, China, Peru, Russia, Australia
Gold	China, Australia, Russia, USA, Canada
Tin	China, Indonesia, Peru, Bolivia, Brazil
Zinc	China, Peru, Australia, USA, Canada
Uranium	Kazakhstan, Canada, Australia, Niger, Namibia

- Energy resources are generally defined as anything that can be used as a source of energy. Some energy resources are coal, petroleum and natural gas etc.

Resources	Major Producer Countries
Coal	China, USA, Australia, India
Petroleum	USA, Saudi Arabia, Russia, China
Natural Gas	USA, Russia, Qatar, Iran
Uranium	Kazakhstan (First), Canada, Australia, Niger, Namibia
Wind Energy	China, USA, Germany, Spain, India
Solar Energy	Germany, China, Italy, Japan, USA, Spain, France
Hydro-electricity	China (First), Canada, Brazil, USA, Norway, New Zealand, India

INDUSTRIES

- All economic activities namely; primary, secondary, tertiary and quaternary, revolve around obtaining and utilising resources necessary for survival.
- Among all, these secondary activities add value to natural resources by transforming raw materials into valuable products.

Industries	Major Countries
Iron and Steel Industry	Russia, China, USA, Japan, India, Britain
Cotton Textile	China, India, USA
Jute Industry	India, Bangladesh, China
Woolen Textile	Australia, China, USA
Paper Industries	China, USA, Germany, Japan
Ship Building	China, South Korea, Japan
Aircraft Building	USA, Britain, Russia, France
Automobile Industry	China, USA, Japan, Germany, South Korea, India
Petro Chemical	USA, Belgium, Japan, Netherlands, China, South Korea, Taiwan

TRANSPORT

Transport along with communication is considered as lifeline of an economy. The principle modes of transportation includes land transport, waterways, airways and pipelines.

Road Transport

- Roadways are the most universal and accessible form of transportation. It is also very economical for short distances.
- Germany was the first country in the world to establish a nation wide highway network.
- USA has world's longest and biggest road network followed by China, India, Brazil and Russia.
- Pan-American Highway is the longest highway in the world.

Rail Transport

- It is preferred mode of land transport for bulky goods passengers over long distances.
- The **first train** was started on 27th September, 1825 between Stockton to Darlington in England.
- United States has world's longest rail network, followed by Russia and China.
- Trans-Siberian Railway is the longest railway line in the world.

Air Transport

- Air transport is the fastest but costliest mode of transport. The development of air transport started after the World War I. But the actual development of this transport took place after the World War II.
- *Three types of services are performed by air transport :*
 - i. Inter-Continental Global** This includes the longest services. e.g. services between the continents.
 - ii. Continental Air Route** These services are performed among the countries within a continent. e.g. India-China.

- iii. **National Air Route** Perform services for long distance within the country. e.g. Mumbai-Delhi.

INTERNATIONAL AIR TRANSPORT AUTHORITY (IATA)

- It is an international trade body created over 60 years ago by a group of airlines. Today, IATA represents 240 airlines comprising 84% of total air traffic.
- The organisation also represents, leads and serves the airline industry in general.

Water Transport

- Both ocean water transport and Inland transport form the part of water transport.
- Waterways are the most suitable and cheapest mode of transport for carrying international trade.

Important Sea Routes of the World

North Atlantic Sea Route (Big Trunk) It links North-Eastern USA and North-Western Europe. It is busiest route in the world.

South Atlantic Sea Route This route connects North America and Europe with South America.

Mediterranean-Indian Ocean Sea Route It connects highly industrialised West European region with East Africa, South Africa, South-East Asia and Australia and New Zealand.

Cape of Good Hope Route It connects West European and West African countries with Brazil, Argentina and Uruguay in South America.

North Pacific Sea Route This route links the ports of West-coast of North America with those of Asia.

South Pacific Sea Route This route is an important connecting link between South and North America with Australia, New Zealand and various Pacific islands *via* Panama Canal.

Major Canal System of the World

Suez Canal It was constructed in 1869. This is the largest canal in the world, that connects Port Said in North and Port Suez in South Linking Red Sea and Mediterranean Sea.

Panama Canal It is constructed across the Panama Isthmus between Panama city and Colon, connecting the Pacific Ocean and the Caribbean Sea.

Kiel Canal This canal in Germany connects the North Sea with the Baltic Sea.

Manchester Shipping Canal It connects Manchester with Eastham.

North-Sea Canal It joins North-sea with Amsterdam.

Inland Waterways

- Inland waterways were the chief mode of transportation before the advent of railways.
- The major oceans of world are connected with each other thus, forming a continuous expanse of water providing links to most parts of the world.

Major Inland Waterways of the World

Rhine Waterways It is navigable for 700 km from Rotterdam (Netherlands) to Basel (Switzerland). This is world's most heavily used waterways.

Danube Waterway Danube river rises in Black Forest and flow Eastward in Eastern Europe. It is navigable upto Taurna Severin.

Volga Waterways It is an important waterways of Russia and is navigable for 11200 km. The Volga-Moscow Canal connects it with Moscow region and Volga-Don canal with Black sea.

Great Lakes-St. Lawrence Seaway The Great lakes of North America, Superior, Hurie, Erie and Ontario are connected by Soo canal and Welland canal. Great Lakes along with estuary of St. Lawrence river form the waterways.

Mississippi Waterways The Mississippi-Ohio waterway connects interior part of USA with Gulf of Mexico.

Races

- Race is a term used for large group of people with some basic inherited physical characteristics common. e.g. skin colour, hair, facial features, eye colour, head shape etc.
- Four major human races are Caucasoid, Australoid, Mongoloid and Negroid.

Trait	Caucasoid	Mongoloid	Negroid	Australoid
Skin colour	Pale reddish white to olive brown	Saffron to yellow brown	Brown to brown-black	Skin colour is of chocolate
Head form	Long to broad and short, medium high to very high	Predominantly broad, height medium	Predominantly long height low to medium	Dolichocephalic
Face	Narrow to medium broad, tends to high, no prognathism	Medium broad to very broad, malars high and flat, tends to medium high	Medium broad to narrow, tends to medium high, strong prognathism	Large heavy jaws and prognathism
Hair	Hair colour, light blonde to dark brown	Hair colour, brown to brown black	Hair colour, brown black	Hair is silky, black and wavy
Eye colour	Light blue to dark brown	Brown to dark brown	Brown to brown black	dark brown
Nose	Bridge usually high	Bridge usually low to medium	Bridge usually low	

Major Tribes of the World

- The term tribe refers to a cultural or social rather than a biological or racial groups.

Tribe	Country/Region
Aborigines	Australia
Apache	South-West America
Ainu	Japan
Amerinds	America
Acta	Philippines
Bantu	Central and Southern America
Beja	Egypt
Berber	Northern coast of Africa
Beduins	Arabia, Syria, North Africa
Bora	Columbia, Brazil
Bushmen	Kalahari Desert in Africa
Buryats	Siberia, Central Asia
Chukchi	Chukchi Peninsula (Russia)
Dayak	Borneo (Indonesia)
Fellah	Middle East, North Africa
Fulani	Nigeria
Han	China
Hottentots	South-West Africa
Hausa	Nigeria
Hadza	Tanzania
Inca	Peru, Mexico
Inuit	Greenland, Canada, Alaska
Jarawa	Andaman (India)
Kamasin	Siberia
Karakalpak	Central Asia
Kirghiz	Kyrgyzstan
Kubu	Sumatra (Indonesia)
Kurds	Turkey, Iran, Iraq, Syria
Kung	Southern Africa
Kamba	Kenya
Kikuyu	Kenya
Lapps	Finland, Sweden, Norway, Russia
Lepcha	Sikkim
Maori	Australia
Papuan	New Guinea
Red Indian	USA
Semang	Malaysia
Shan	Myanmar
Swahili	Kenya, Tanzania
Tatars	Mongolian Plateau
Tungus	Siberia
Vedda	Sri Lanka
Yuit	Alaska, Siberia
Zulu	South Africa

Population

Population refers to the whole number of people living in country or region. Demography is the scientific study of human population.

Population Distribution

- Population is not distributed evenly in all parts of the world. Some places have sparse (scattered) population and some have dense population.
- Sparsely populated places tend to be difficult places to live. These are usually places with hostile environment e.g. desert areas of the world.
- More than 75% of the world's population lives in the continents of Africa and Asia.

	Country	Population (million)		Country	Population (million)
1.	China	1369	2.	India	1270
3.	USA	321	4.	Indonesia	255
5.	Brazil	204	6.	Pakistan	189
7.	Nigeria	183	8.	Bangladesh	158
9.	Russia	146	10.	Japan	126

Population Density

- Density is defined as number of persons in per unit area. The density varies not only from one country to another within a continent but also from one part to the other within a country.
- The highest densities are found in North- Eastern USA, North-West Europe, South Asia, South-East Asia and East Asia.
- The lowest densities are found in hot and cold deserts, areas around pole, high rainfall area near equator.

Demographic Variables

Birth Rate Number of live births per 1000 population in a given year. High birth rate found in Africa while lowest is in Europe.

Death Rate Number of deaths per 1000 population in a given year.

Infant Mortality Rate The number of deaths of infants under per 1000 live births in a year.

Maternal Mortality Rate The number of women who die as a result of pregnancy and child birth complications per 100000 live births in a year.

Total Fertility Rate The average number of children that would be born alive to a women.

Population Pyramid A bar chart that is arranged vertically to show the distribution of population by age and sex. Males are on the left and females on the right.

Doubling Time The number of years required for the population of an area to double itself, at current rate of population growth.

Age-Dependency Ratio The ratio of persons in the ages defined as dependent (under 15 years and over 64 years) to persons in the ages defined as economically productive (15-64 years).

Demographic Transition It refers to shift of birth and death rates from high to low levels. The decline in fertility is always delayed leading to population growth.

Migration

Migration means the movement of people from one place to the other.

Types of Migration

- In the context of domestic migration, migration can be either in-migration or out-migration.
 - **In-migration** The process of entering one administrative sub-division of a country from another sub-division.
 - **Out-migration** The process of leaving one sub-division of a country to take up residence in another.
- *International migration can be classified into the following :*
 - **Emigration** It refers to the process of leaving one country to take up permanent or semi-permanent residence in another.
 - **Immigration** It is a process of entering one country from another to take up permanent or semi-permanent residence.
- **Refugee** A person who is residing outside country of his/her origin due to fear of persecution for reasons of race, religion, nationality or political opinion.

- **Internally Displaced Person** A person who is forced to leave his/her home region due to unfavourable conditions, but does not cross any boundary.

Human Development

- Development means qualitative change, which is always value positive.
- The concept of human development was introduced by Dr Mahbub-ul-Haq. Dr Haq has described human development as that development that enlarges people choices and improves their lives. The four pillars of human development are equality, sustainability, productivity and empowerment.

Human Development Index

- The Human Development Report is an annual milestone publication by the Human Development Report office of the United Nations Development Programme (UNDP).
- The Report was first launched in 1990 by the world renowned Pakistani economist Mahbub-ul-Haq and Indian Nobel laureate Amartya Sen.
- Human development is measured in terms of life expectancy at birth, education and per capita income.
- Norway has highest HDI ranking. India ranks 130th in the list.
 - (i) HDi ranking (2016)
 - (ii) India ranks 131
 - (iii) Norway topped the Index

> PRACTICE EXERCISE

1. Which of the following is the chief characteristic of 'mixed farming'?

- (a) Cultivation of both cash crops and food crops
- (b) Cultivation of two or more crops in the same field
- (c) Rearing of animals and cultivation of crops together
- (d) None of the above

2. Consider the following statements

- 1. Areas with high density of population have generally longer jhum cycle.
- 2. In New Zealand, cattle are kept on fairly small farms and carefully fed to give quality meat.
- 3. Cash crops are never grown in jhuming cultivation.
- 4. Livestock farming is usually a response to difficult conditions.

Which of the statements given above are incorrect?

- (a) 1 and 3
- (b) 2 and 4
- (c) 2 and 3
- (d) 3 and 4

3. Match the following

List I	List II
A. Ranchos	1. India
B. Favelas	2. Brazil
C. Kavettits	3. Venezuela
D. Bustee	4. Myanmar

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 2 3 4 1 | (b) 3 2 4 1 |
| (c) 4 3 2 1 | (d) 4 2 3 1 |

4. Infant mortality rate

- (a) the ratio of number deaths of children under 3 months to the number of live births
- (b) the ratio of number deaths of children under 6 months to the number of live births
- (c) the ratio of number deaths of children under 1 year to the number of live births
- (d) the ratio of number deaths of children under 1 year to the number of children below 5 years

5. Which of the following is a low density area?

- 1. Equatorial forest
- 2. Tropical deserts
- 3. Eastern Asia

4. North-Eastern America

Select the answer using the codes given below.

- (a) 1, 2 and 4
- (b) 1 and 4
- (c) 2 and 3
- (d) 1, 3 and 4

6. Consider the following countries

- 1. Bangladesh
- 2. Brazil
- 3. Indonesia
- 4. Japan

The correct sequence of these countries in descending order of their population is

- (a) 1, 2, 3, 4
- (b) 2, 1, 4, 3
- (c) 3, 2, 4, 1
- (d) 4, 3, 2, 1

7. Match the following

List I	List II
A. Masai	1. North America
B. Berbers	2. East Africa
C. Red Indians	3. Asia
D. Kirghiz	4. North-West Africa

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 3 2 1 4 | (b) 2 3 4 1 |
| (c) 2 4 1 3 | (d) 1 4 2 3 |

8. Match the following

List I (Russian Regions)	List II (Industrials)
A. Central Industrial	1. Agro-based
B. Grozny	2. Cotton textiles
C. Volga	3. Machinery
D. Urals	4. Metallurgy

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 2 4 1 3 | (b) 1 3 5 4 |
| (c) 2 3 1 4 | (d) 2 1 4 3 |

9. Consider the following mangrove areas

- 1. Bhitarkanika
- 2. Pichavaram
- 3. Coondapur

Which among the above is/are situated on the Eastern Coast of India?

- (a) Only 1
- (b) 1 and 2
- (c) 1 and 3
- (d) All of these

10. Match the following

List I	List II
A. Iron and steel	1. Dortmund
B. Automobile	2. Seattle
C. Ship building	3. Detroit
D. Aircraft	4. Portsmouth

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 2 1 4 3 | (b) 2 3 1 4 |
| (c) 1 3 4 2 | (d) 1 4 3 2 |

11. Match the following

List I (Islands)	List II (Countries)
A. Corsica	1. Italy
B. Sardinia	2. Greece
C. Crete	3. France
D. Balearic Island	4. Spain

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 3 1 2 4 | (b) 3 2 1 4 |
| (c) 3 4 1 2 | (d) 3 2 4 1 |

12. Match the following

List I (Mines)	List II (Minerals)
A. Flin Flon	1. Coal
B. Sudbury region	2. Petroleum
C. Kuzbass	3. Copper and Tin
D. Baku	4. Gold

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 1 3 2 4 | (b) 4 3 1 2 |
| (c) 1 2 3 4 | (d) 3 4 1 2 |

13. Match the following

List I	List II
A. Ivanovo	1. Cotton textile industry
B. Magnitogorsk	2. Engineering industry
C. Moscow	3. Iron and steel industry
D. St Petersburg	4. Ship-building industry

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 4 3 2 1 | (b) 1 3 2 4 |
| (c) 1 2 3 4 | (d) 3 2 1 4 |

14. Which water separates Australia from New Zealand?

- (a) Cook straits
- (b) Megallan
- (c) Tasman sea
- (d) Great Barrier Reef

15. The reasons for under exploitation of tropical forests are

- 1. few transport routes.
- 2. mixed species forest.
- 3. willingness of governments to conserve forests.
- 4. the paper made of hardwood is usually costly.

Select the answer using the codes given below.

- (a) 1, 2 and 3
- (b) 1 and 2
- (c) 1, 2 and 4
- (d) All of these